

Education & Certifications	Indian Institute of Technology, Kanpur (2019-24)			BS + MS (Dual Degree)	CGPA: 9.3/10.0
	Financial Derivatives & Quantitative Analysis (C. Connell)			Udemy	2025
	Neuro Match Academy: Foundational Models			NMA Summer Camp	2024
Languages	C++, Python, SQL, VBA	Utilities	Pandas, NumPy, Scikit-learn, Tensorflow, Excel, PowerBI, Tableau		
Coursework	Probability and Statistics Microeconomics Macroeconomics Econometrics Financial Econometrics				
	Calculus & Linear Algebra Differential Equations Advanced Machine Learning				
Competencies	Statistical Modelling Time Series Analysis Machine Learning Image Processing Financial Modelling				

Professional Experience

BUILTXSDC | Founder's Office Aug '24 — Present

- Financial Modelling (Data Analysis) | Steel & Cement Procurement Strategy**
- Modeled **20+ years** of steel and cement **pricing data** using multivariate time series approaches (**ARIMA, ARIMAX, LSTM**), incorporating **macroeconomic** and weather-linked variables (imports, CPI, rainfall trends) discovering seasonal patterns
 - Reduced procurement cost volatility by **5%** via simulated **hedging strategies** using forecast-driven stress tests and downside **risk buffers**. Developed scenario-based **pricing models** to anticipate adverse market regimes, aiding in contract timing.
- Product Management & Marketing (SaaS/B2B) | Construction Marketplace MVP Go-to-Market**
- Analyzed **600 K+ Play Store reviews** with **NLP (VADER sentiment analysis + BERT topic modeling)**, built a weighted pain-point score for backlog ranking. Validated insights through **100+** user interviews/surveys across the user journey.
 - Designed landing pages applying user psychology on page's UX/UI elements with **A/B testings** and iterations.
 - Conducted **Landscape analysis** of India's construction material/subcontractor market, mapping supply chains, digital adoption trends, existing solution analysis; Collected data through stakeholder's surveys, gov. reports & media publications, automated webcrawling.

SCHLUMBERGER (SLB) | Field Engineer Intern May '23 — July '23

Operational Efficiency & Risk Management	- Managed an inventory of 500+ accessories/tools, reorganizing its flow between the labs and the warehouse - Created a mail alert system to ensure stock levels of MNS and assets remain above critical thresholds - Strengthened safety management by filing 25+ RIRs and walk-around reports for hazard evaluation - Crafted a Universal Standard Operating Procedure for remote operations reducing operators from 10 to 3 - Designed Site Operations Flowcharts and created Rig Network Contingency Protocols used in 100+ jobs/yr
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CONNECTEDH | Data Analyst Intern Jun '21 — Sep '21

Diagnostic Centre Data	- Analyzed appt scheduling data, creating a dashboard to track wait times and sample collection efficiency - Recommended staffing & process improvements, to reduce wait times & enhance home collection services
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Machine Learning Projects

MASTER'S THESIS | Water Quality mapping from UAV images using ML Aug '23 — May '24

- Author of AGU 2024 abstract: [link](#) | Attained A* in all thesis credits
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| Metals Conc. Prediction: Image Mapping | <ul style="list-style-type: none">Created lab-generated Heavy metals solution spectrum to feed predictive model and used as ground truthApplied derivative spectroscopy and numerical processing, developed WQ maps using python Spy libraryCategorized metals on their optical behaviors correlation using Hierarchical Ward's statistical clusteringApplied PSLR and OSLR models on reflectance data to predict concentrations of Metals achieving R² 0.68 |
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NATIONAL UNIVERSITY of SINGAPORE (NUS) | Data Summer Intern May '22 — July '22

Time Series: Image Modelling	- Detected and rectified missing vegetation data due to cloud coverage at 22 River sites in South India - Examined Satellite SAR images for penetration and Earth Engine's log-scaled pre-processing for visualization - Forecasted vegetation dynamics using MODIS NDVI time series data using ML techniques like LSTM/RNN - Estimated the missing normalized difference vegetation index from SAR images dataset with R² 0.84
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Position of Responsibility

FOUNDER - OVERALL COORDINATOR | Sustainability Cell, IITK Sep '23 — Apr'24

Leadership	- Led the establishment of the pioneering student-led 'Sustainability Cell' for 9k+ campus residents - Managed team of 200+ volunteers, oversaw entire logistics to execute the Institute's first cleanliness drive
Achievements	- Partnered in key project: 'IITK towards carbon neutral', institute's mission of reducing its carbon footprint - Organized India's largest Climate hackathon in collab' with BuzzOnEarth seeing 1k+ teams' participation

STUDENT REPESENTATIVE | UG Committee, IITK Sep '21 — Aug '23

- Advocated** for students' interests, termination, graduation eligibility in the committee. Led extensive **student surveys**. Mentored **200+** dep' students. Facilitated departmental **audit meetings** attended by likes of Prof. W. White (ESAC, NASA)
- Tutored offline-class Thermodynamics to **400+** UG students as Teaching Assistant; Worked in a 3-tier team to raise **INR 2.2 cr** supporting **441** underprivileged under "Covid-19 Relief Fund"